DSA Assignment 5

Name : Nandini Ravindra Chimandare
Batch/Div : SY AIML B1
PRN : 202501110108

1. Aim

To implement a Stack using two Queues in C++ and perform basic stack operations such as Push, Pop, Top, and Display.

2. Theory

A Stack follows the LIFO (Last In, First Out) principle, which means the element inserted last is removed first.

A Queue follows the FIFO (First In, First Out) principle, which means the element inserted first is removed first.

In this program, we use two queues, q1 and q2, to implement stack operations. Whenever we insert a new element, we arrange the queues so that the newest element stays at the front of q1.

3. Algorithm

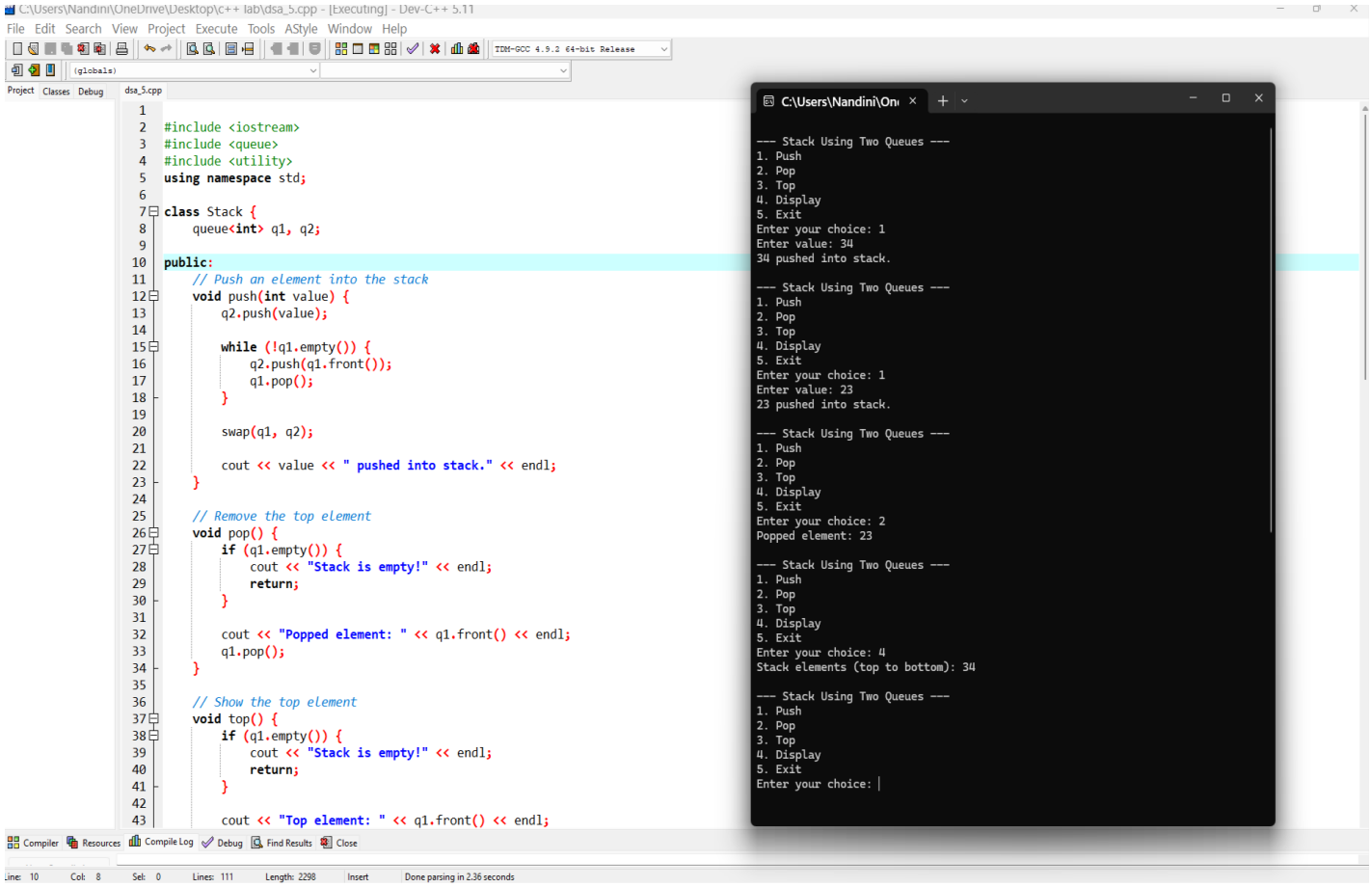
1. Start the program and create two queues, q1 and q2.
2. For Push, insert the new element into q2.
3. Move all elements from q1 to q2.
4. Swap q1 and q2, so the newest element comes first.
5. For Pop, remove the front element of q1.

6. For Top, display the front element of q1 without removing it.

7. For Display, print all elements of q1.

8. Repeat the operations using a menu until the user selects Exit.

Implementation:



```
#include <iostream>
#include <queue>
#include <utility>
using namespace std;

class Stack {
    queue<int> q1, q2;

public:
    // Push an element into the stack
    void push(int value) {
        q2.push(value);

        while (!q1.empty()) {
            q2.push(q1.front());
            q1.pop();
        }

        swap(q1, q2);

        cout << value << " pushed into stack." << endl;
    }

    // Remove the top element
    void pop() {
        if (q1.empty()) {
            cout << "Stack is empty!" << endl;
            return;
        }

        cout << "Popped element: " << q1.front() << endl;
        q1.pop();
    }

    // Show the top element
    void top() {
        if (q1.empty()) {
            cout << "Stack is empty!" << endl;
            return;
        }

        cout << "Top element: " << q1.front() << endl;
```

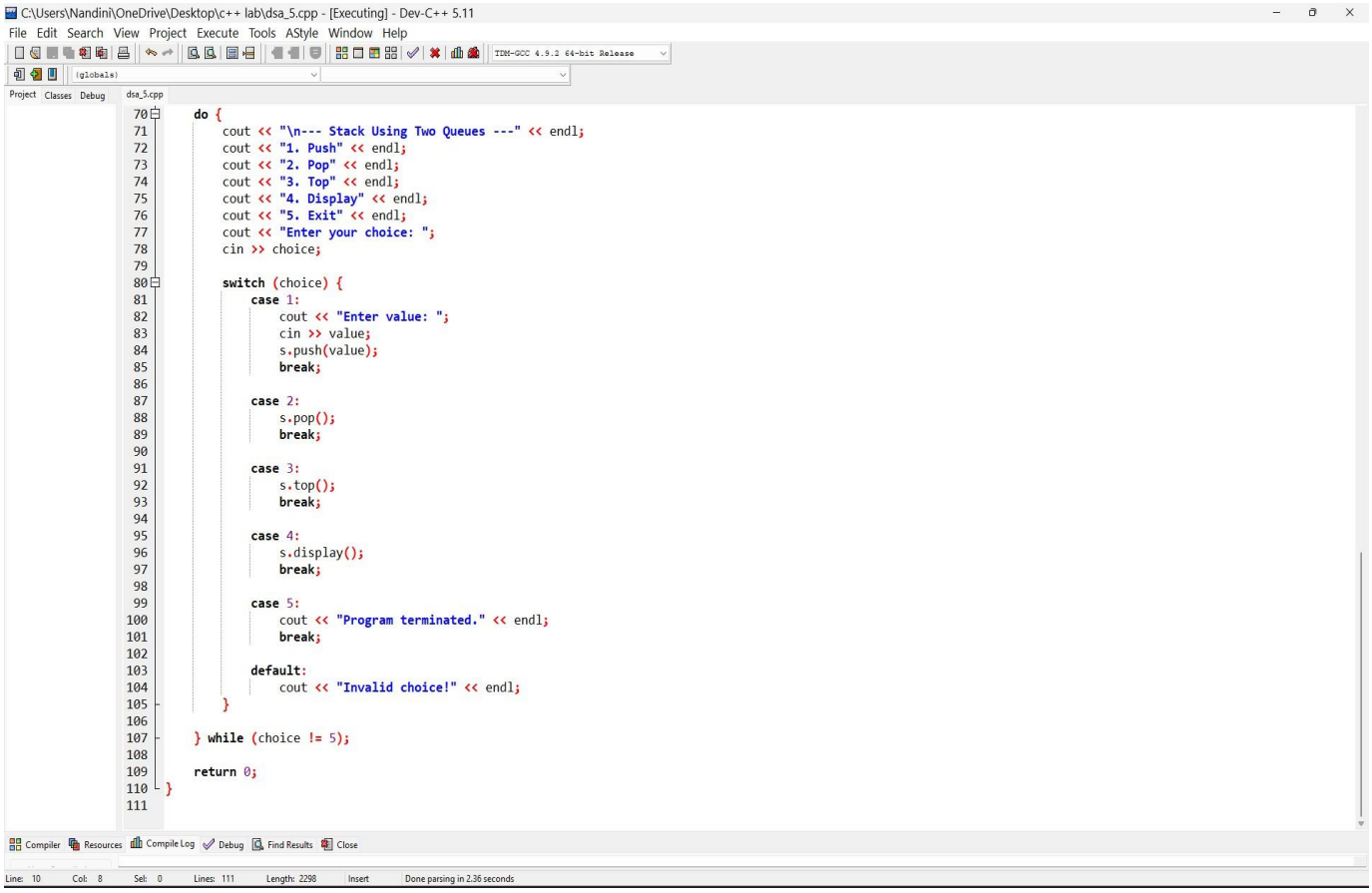
```
--- Stack Using Two Queues ---
1. Push
2. Pop
3. Top
4. Display
5. Exit
Enter your choice: 1
Enter value: 34
34 pushed into stack.

--- Stack Using Two Queues ---
1. Push
2. Pop
3. Top
4. Display
5. Exit
Enter your choice: 1
Enter value: 23
23 pushed into stack.

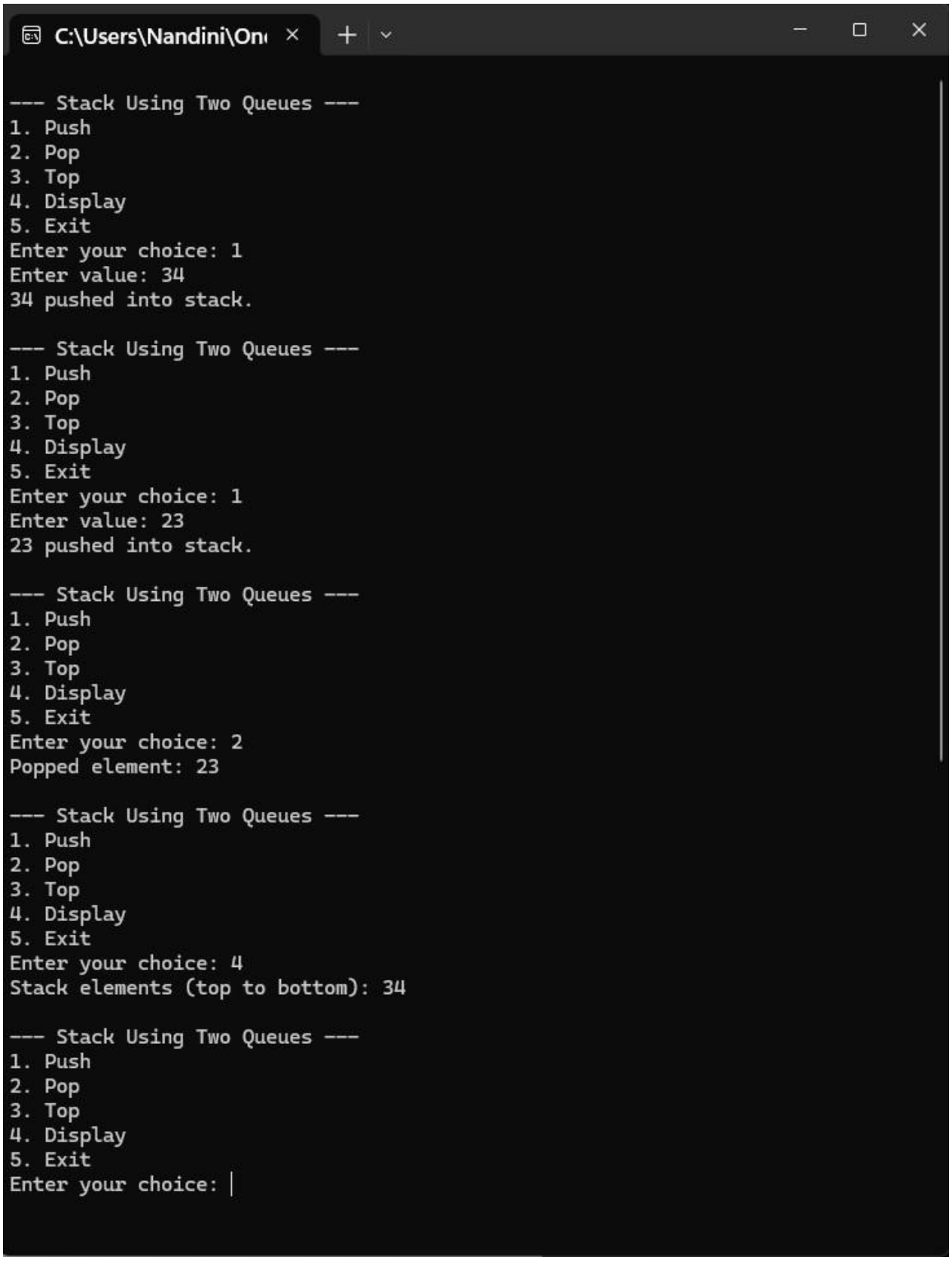
--- Stack Using Two Queues ---
1. Push
2. Pop
3. Top
4. Display
5. Exit
Enter your choice: 2
Popped element: 23

--- Stack Using Two Queues ---
1. Push
2. Pop
3. Top
4. Display
5. Exit
Enter your choice: 4
Stack elements (top to bottom): 34

--- Stack Using Two Queues ---
1. Push
2. Pop
3. Top
4. Display
5. Exit
Enter your choice:
```


C:\Users\Nandini\OneDrive\Desktop\c++ lab\dsa_5.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
dsa_5.cpp
do {
cout << "\n--- Stack Using Two Queues ---" << endl;
cout << "1. Push" << endl;
cout << "2. Pop" << endl;
cout << "3. Top" << endl;
cout << "4. Display" << endl;
cout << "5. Exit" << endl;
cout << "Enter your choice: ";
cin >> choice;
switch (choice) {
case 1:
cout << "Enter value: ";
cin >> value;
s.push(value);
break;
case 2:
s.pop();
break;
case 3:
s.top();
break;
case 4:
s.display();
break;
case 5:
cout << "Program terminated." << endl;
break;
default:
cout << "Invalid choice!" << endl;
}
} while (choice != 5);
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Line: 10 Col: 8 Sel: 0 Lines: 111 Length: 2298 Insert Done parsing in 2.36 seconds

Output:



6. Time and Space Complexity

Operation	Time complexity
Push	(O(n))
Pop	(O(1))
Top	(O(1))
Display	(O(n))

Space complexity: (O(n)), because the queues store the stack elements.

7. Conclusion

Thus, a Stack was successfully implemented using two Queues in C++. The program performs Push, Pop, Top, and Display operations and follows the LIFO principle**.**